EvidenceFirst: Pre-Generation Evidence-Chain Verification for Traceable KG-RAG

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Abstract. Knowledge graph-enhanced retrieval-augmented generation (KG-RAG) is a natural fit for multi-hop question answering because graph structure can expose the entity and relation chain needed to answer a question. Yet most KG-RAG systems still verify too late: they retrieve a graph neighborhood, ask a language model to generate an answer, and only then evaluate the final string. When the retrieved graph is incomplete, the system may still produce a fluent answer but cannot say whether the failure came from a missing entity, a broken bridge relation, a disconnected component, or the generator itself.
We present EvidenceFirst, a training-free KG-RAG pipeline that verifies evidence before generation. The central object of verification is not the answer string but the graph-structured evidence chain. EvidenceFirst builds a per-question evidence graph, checks whether the graph contains a complete multi-hop chain, diagnoses graph gaps, performs localized repair, and then generates a grounded answer from the verified or repaired evidence. The output is therefore both an answer and an audit trail: chain completeness, missing entities, disconnected pairs, repair attempts, fallback cases, and runtime failures.
This draft reports the completed comparison under a strict reporting boundary. On HotpotQA-1000, the only reported EvidenceFirst main model is final v6, which reaches 56.2% EM and 65.6% F1, exceeding HopRAG strict by 2.1 EM and 1.26 F1 points. On 2WikiMultiHopQA (500 questions), EvidenceFirst v6 reaches 69.0% EM and 75.8% F1, obtaining the best EM among completed runs; LightRAG remains slightly higher in F1. HopRAG strict for 2Wiki is still running, and MS GraphRAG for 2Wiki has not been run.

Keywords: KG-RAG · GraphRAG · Evidence Verification · Multi-Hop QA · HotpotQA · 2WikiMultiHopQA · Traceable Reasoning

1 Introduction

Multi-hop question answering over Web knowledge requires more than retrieving a passage that shares surface words with the question. A correct answer often depends on a chain of intermediate entities and relations: an entity mentioned in the question leads to a bridge page, the bridge page identifies another entity, and only then can the answer be derived. Retrieval-augmented generation (RAG) grounds generation in retrieved text [4]; KG-RAG and GraphRAG systems add structured entity-relation evidence over the retrieved corpus [1,2,3]. Graphs are a natural fit for multi-hop reasoning because the answer is often the endpoint of a path through evidence.

The open problem is not only retrieval quality. It is evidence sufficiency. A KG-RAG system may retrieve several relevant triples and still miss the bridge edge that makes the answer derivable. It may retrieve both endpoint entities but leave them in disconnected components. It may retrieve a graph neighborhood whose entities look plausible but do not form the required reasoning chain. In all of these cases, a retrieve-then-generate pipeline can still produce a confident answer. The final answer string may be evaluated later, but the system has already lost the ability to explain whether the failure was caused by retrieval, graph construction, evidence connectivity, or generation.

EvidenceFirst addresses this problem by making the evidence graph the verified object. Before generation, the system checks whether the retrieved graph contains a complete multi-hop evidence chain for the question. If the chain is incomplete, EvidenceFirst emits a graph-structural diagnostic: which entity is missing, which endpoints are disconnected, whether a bridge edge is absent, or whether evidence retrieval failed entirely. The diagnostic then drives targeted repair, after which the generator receives the verified or repaired evidence.

Contributions. This paper makes three contributions.

- We introduce pre-generation evidence-chain verification for KG-RAG: the verified object is the retrieved evidence path, not only the generated answer.
- We define graph-structural diagnostics for multi-hop QA, including chain completeness, missing entities, disconnected pairs, bridge gaps, and fallback cases.
- We evaluate EvidenceFirst on HotpotQA-1000 and 2WikiMultiHopQA-500 against Naive RAG, IRCoT, A-RAG, LightRAG, MS GraphRAG, and HopRAG where completed, and report controlled ablations that separate answer-quality gains from diagnostic traceability.

2 Method

2.1 Pipeline Overview

Let q be a multi-hop question, $\mathcal{P} = \{p_i\}$ a set of candidate passages, and $\mathcal{G} = (V, E)$ a per-question evidence graph extracted from the passages. EvidenceFirst uses six stages:

1. build or load a per-question KG from candidate passages;
2. retrieve KG triples and top-ranked passages for the question;
3. construct an evidence graph from retrieved triples;
4. check graph-structural chain completeness;
5. repair localized graph gaps when possible;
6. generate a concise answer from verified or repaired evidence.

Algorithm 1 summarizes inference.

Algorithm 1 EVIDENCEFIRST inference

Require: Question q, passages $\mathcal{P}$, per-question KG $\mathcal{G}$
Ensure: Answer a, diagnostic record D
$D \leftarrow \emptyset$
$T \leftarrow \text{retrieve\_triples}(q, \mathcal{G})$
if $T = \emptyset$ **then**
 $D \leftarrow D \cup \{\text{EMPTYEVIDENCE}\}$
 $P_q \leftarrow \text{local\_passages}(q, \mathcal{P})$
 $a \leftarrow \text{passage\_read}(q, P_q, D)$
else
 $G_E \leftarrow \text{build\_evidence\_graph}(T)$
 $(\mathit{complete}, D) \leftarrow \text{check\_chain\_completeness}(q, G_E)$
 if not *complete* **then**
  $(T', D_r) \leftarrow \text{repair\_gap}(q, \mathcal{G}, D)$
  $T \leftarrow T \cup T'$
  $G_E \leftarrow \text{build\_evidence\_graph}(T)$
  $(\mathit{complete}, D') \leftarrow \text{check\_chain\_completeness}(q, G_E)$
  $D \leftarrow D \cup D_r \cup D'$
 end if
 $P_q \leftarrow \text{local\_passages}(q, \mathcal{P})$
 $a \leftarrow \text{grounded\_generate}(q, T, P_q, D)$
end if
$a \leftarrow \text{answer\_canonicalize}(q, a, \mathcal{P})$
return (a, D)

Table 1. Graph-structural diagnostics emitted by EVIDENCEFIRST before generation.

Gap type	Graph condition	Typical repair action
Missing entity	Required node absent	Entity expansion / passage recall
Disconnected pair	Endpoints in separate components	Bridge search
Bridge gap	Partial path without bridge edge	Targeted relation retrieval
Empty evidence	No usable triples retrieved	Passage fallback

2.2 Evidence-Chain Diagnostics

The verifier returns structured diagnostics rather than a single score. Table 1 lists the main diagnostic categories. These labels serve two roles: they guide localized repair during inference, and they provide an audit trail for posthoc error analysis.

The completeness check is structural rather than oracle-semantic. A chain is treated as complete when the evidence graph contains the question entities and a connected support structure linking them through retrieved triples or repaired evidence. A complete chain can still be misread by the generator, but an incomplete chain warns that the system should repair or fallback before answer generation.

3 Experimental Setup

Datasets. We report two evaluation settings. HotpotQA-1000 is a 1000-question distractor-style multi-hop QA set with repaired contexts from the HotpotQA validation split. 2WikiMultiHopQA-500 is a 500-question sample whose examples contain natural-language contexts together with Wikidata-derived evidence triples, entity ids, evidence ids, and answer ids. Each 2Wiki example includes 10 context passages; the 500 examples expand to 5000 question-context pairs and 3346 unique passage bodies for graph-based baselines.

Baselines. For HotpotQA-1000, completed baselines are Naive RAG, IRCoT, A-RAG, LightRAG, MS GraphRAG, and HopRAG strict response-only rescoring. For 2Wiki-500, completed baselines are Naive RAG, IRCoT, A-RAG, and LightRAG. HopRAG strict for 2Wiki is running at the time of this draft; MS GraphRAG for 2Wiki was intentionally not run after the user decision to defer it.

Metrics. We use exact match (EM) and token F1 with standard QA normalization: lowercasing, article removal, punctuation removal, and whitespace collapse. We report full-sample results only in the main tables. Partial smoke results are excluded from main comparisons.

Reporting boundary. For HotpotQA-1000, we report only the final EvidenceFirst v6 main model. Earlier HotpotQA development variants are treated as process artifacts and are omitted from the main comparison. Controlled ablations use explicit pipeline switches and are reported separately from historical development variants.

Artifact mapping. All main numbers in this draft are copied from repository artifacts. The HotpotQA baseline summary is `results/wise_hotpot1000_baseline_summary.csv`. The HotpotQA EvidenceFirst v6 row uses `experiments/evidence_first/results/hotpotqa_evidence_fi` The 2Wiki baseline rows use `results/wise_2wiki_naive_rag_metrics.csv`, `results/wise_2wiki_ircot_metrics.csv`, `results/wise_2wiki_arag_full_post_metrics.csv`, and `results/wise_2wiki_lightrag_metrics.csv`. The 2Wiki EvidenceFirst row uses `results/wise_2wiki_evidencefirst_v6_readerfull_canon_metrics.csv`. The controlled ablation rows use the 2026-06-08 artifacts with prefix `experiments/evidence_first/results/*e`

4 Results

4.1 HotpotQA-1000

Table 2 reports the completed HotpotQA-1000 baseline comparison together with the final EvidenceFirst v6 main model. EvidenceFirst v6 reaches 56.2% EM and 65.6% F1. This exceeds HopRAG strict by 2.1 EM and 1.26 F1 points. Other HotpotQA development attempts are excluded from this main comparison.

The result supports a conservative claim: pre-generation evidence-chain verification can remain competitive with strong graph baselines while adding diagnostic

Table 2. HotpotQA-1000 results. All rows are complete 1000-question runs.

Method	N	EM	F1
Naive RAG	1000	0.3920	0.4865
IRCoT	1000	0.4660	0.5707
A-RAG	1000	0.5170	0.6273
LightRAG	1000	0.4380	0.5376
MS GraphRAG	1000	0.1890	0.2774
HopRAG strict	1000	0.5410	0.6438
EvidenceFirst v6	1000	**0.5620**	**0.6564**

Table 3. HotpotQA-1000 question-type breakdown against the strongest completed graph baselines.

Type	N	A-RAG		HopRAG strict		EvidenceFirst v6	
		EM	F1	EM	F1	EM	F1
Bridge	500	0.3500	0.4893	0.3980	0.5221	**0.4560**	**0.5689**
Comparison	500	0.6840	0.7652	**0.6840**	**0.7654**	0.6680	0.7439

traceability. The HotpotQA claim is intentionally limited to final v6, rather than intermediate attempts explored during development.

Table 3 shows where the HotpotQA gain comes from. Against the strongest completed graph baselines, EvidenceFirst is strongest on bridge questions: it improves over HopRAG strict by 5.8 EM points and over A-RAG by 10.6 EM points. On comparison questions, A-RAG and HopRAG strict remain stronger. This is consistent with the method design: evidence-chain verification is most useful when the answer depends on connecting entities across evidence, while attribute-comparison questions can often be solved by direct answer comparison once the relevant passages are retrieved.

4.2 2WikiMultiHopQA-500

Table 4 reports completed 2Wiki-500 results. EvidenceFirst v6 reader-full canonicalization obtains the highest EM among completed systems, improving over LightRAG by 1.6 EM points, over A-RAG by 2.4 EM points, and over IRCoT by 3.4 EM points. LightRAG remains slightly stronger in F1, exceeding EvidenceFirst by 1.35 F1 points. This indicates that EvidenceFirst produces more exact canonical answers, while LightRAG often obtains partial token overlap even when exact canonicalization differs.

The 2Wiki setting is more favorable for evidence-chain verification than plain answer-only QA because examples include gold evidence triples and entity ids. The current result is answer-level only; future revisions should also evaluate whether the predicted evidence chain matches the annotated Wikidata-derived relation chain.

Table 4. 2WikiMultiHopQA-500 results. HopRAG strict is still running and is therefore not included as a completed row.

Method	N	EM	F1
Naive RAG	500	0.2980	0.3338
IRCoT	500	0.6560	0.7301
A-RAG	500	0.6660	0.7323
LightRAG	500	0.6740	**0.7716**
EvidenceFirst v6	500	**0.6900**	0.7581

Table 5. Controlled EvidenceFirst ablations. Full rows use final v6; ablations use the 2026-06-08 explicit ablation switches.

Variant	HotpotQA-1000		2Wiki-500	
	EM	F1	EM	F1
Full EvidenceFirst v6	0.5620	0.6564	0.6900	0.7581
w/o evidence-chain verification	0.5680	0.6647	0.6780	0.7579
w/o graph repair	0.5540	0.6580	0.6500	0.7424
w/o full/local reader context	0.2360	0.2899	0.1840	0.2289
w/o answer refinement	0.5510	0.6485	0.6900	0.7782

4.3 Component Ablations

Table 5 reports controlled component ablations on both datasets. The results separate components that directly drive answer quality from components that primarily provide traceability and failure diagnosis. Removing local/full reader context causes the largest drop on both datasets, showing that graph evidence alone is not sufficient for robust answer surface realization. Removing answer refinement hurts HotpotQA but increases 2Wiki F1 while preserving 2Wiki EM, suggesting that canonical answer selection can improve exact normalization while reducing partial token overlap. Verification and repair show mixed answer-level effects: repair improves EM relative to the no-repair setting, but verification does not monotonically improve EM/F1 in the current implementation. We therefore interpret these ablations as behavioral evidence, not as a claim that every diagnostic stage independently increases final-answer accuracy.

4.4 Selector Calibration

Table 6 analyzes the final answer-selection signal inside EvidenceFirst v6. The selector is not a standalone accuracy guarantee, and the ablation in Table 5 shows that answer refinement interacts with dataset-specific EM/F1 normalization. However, the selected cases are much more accurate than non-selected cases on both datasets. This supports a narrower claim: answer selection is a useful calibration signal for auditing when the pipeline trusts the refined evidence-grounded answer.

Table 6. EvidenceFirst v6 selector calibration. "Selected" means that the final evidence-grounded postprocess answer was chosen.

Dataset	Selected?	N	Rate	EM	F1
HotpotQA-1000	Yes	816	0.816	**0.6250**	**0.7346**
HotpotQA-1000	No	184	0.184	0.2826	0.3096
2Wiki-500	Yes	352	0.704	**0.7358**	**0.8017**
2Wiki-500	No	148	0.296	0.5811	0.6546

Table 7. Completed EvidenceFirst variants on 2Wiki-500.

Variant	N	EM	F1
v6 default	500	0.3920	0.4582
v6 fresh KG CPU	500	0.3940	0.4565
v6 local context 2	500	0.5040	0.5945
v6 reader-full	500	0.6540	0.7470
v6 reader-full canonical	500	**0.6900**	**0.7581**

The 2Wiki canonicalizer is also conservative: it changes only 18 of 500 predictions, and all 18 changed predictions are exact matches. We therefore use this analysis as supporting evidence for auditability and normalization, not as a replacement for the controlled answer-refinement ablation.

4.5 2Wiki Variants

Table 7 reports the completed EvidenceFirst variants on 2Wiki-500. The low scores for the fresh-KG CPU and local-context-only variants show that answer quality depends strongly on letting the reader see the full local context. The reader-full canonical variant is the current best 2Wiki EvidenceFirst artifact.

4.6 2Wiki Question-Type Analysis

Table 8 compares EvidenceFirst v6 and LightRAG by 2Wiki question type. EvidenceFirst gains most clearly on pure comparison questions. LightRAG remains stronger on bridge-comparison F1 and inference F1, suggesting that EvidenceFirst's canonical answer selection helps exact matching but can under-preserve partial answer overlap.

4.7 2Wiki Evidence Availability

Because 2Wiki provides annotated support facts and evidence triples, we also run an offline evidence-availability analysis. This analysis does not call an LLM. It asks whether the evidence visible to the answer reader contains the gold support titles and evidence entities. Table 9 compares EvidenceFirst v6 reader-full against deterministic or logged evidence views available for Naive RAG and

Table 8. 2Wiki-500 question-type breakdown for LightRAG and EvidenceFirst v6.

Type	N	LightRAG		EvidenceFirst v6	
		EM	F1	EM	F1
Bridge comparison	112	0.9464	0.9589	0.9375	0.9375
Comparison	119	0.8992	0.9034	**0.9580**	**0.9652**
Compositional	218	0.4450	**0.6041**	**0.4541**	0.5626
Inference	51	0.5294	**0.7686**	0.5294	0.7167

Table 9. 2Wiki-500 evidence availability. A-RAG actual read uses logged `chunks_read_ids`; A-RAG search upper bound uses all logged keyword-search hits.

Evidence view	Support-title recall	Evidence-entity coverage
EvidenceFirst v6 reader-full input	**1.0000**	**0.9468**
Naive BM25 top-5 input	0.6020	0.6685
A-RAG actual read chunks	0.2955	0.3326
A-RAG keyword-search upper bound	0.5885	0.6085

A-RAG. LightRAG is not included in this table because the completed LightRAG artifacts do not expose comparable per-query retrieved support titles.

The result supports a narrow evidence-availability claim: EvidenceFirst v6 does not filter away annotated 2Wiki support passages before answer generation, whereas BM25 top-5 and A-RAG's actually read chunks often expose only part of the gold support set. The same offline analysis also shows diagnostic selectivity: chain-complete cases reach 70.29% EM and 76.81% F1, while disconnected-gap cases fall to 45.45% EM and 53.51% F1 and short-chain cases fall to 63.64% EM and 70.35% F1. We therefore use this table as auxiliary evidence for traceability and evidence availability, not as a full retrieval leaderboard.

4.8 Pending Baselines

Table 10 records the status of runs that are not yet part of the completed comparison. These entries should not be used for claims until their full metrics are available.

5 Discussion

Accuracy is competitive, but traceability is the main claim. On HotpotQA-1000, EvidenceFirst improves over HopRAG strict while adding graph-level diagnostics. On 2Wiki-500, EvidenceFirst obtains the best completed EM, although LightRAG remains higher in F1. The robust claim is therefore not an unqualified leaderboard claim; it is that evidence-chain verification can remain competitive while making retrieval failures inspectable.

Table 10. Pending or deferred 2Wiki-500 baselines.

Method	Status	Notes
HopRAG strict	Running	Generation is in progress after graph construction.
MS GraphRAG	Deferred	User decision: do not run for now.

Question-type and selector analyses sharpen the claim. The HotpotQA gain is concentrated on bridge questions, where EvidenceFirst exceeds the strongest graph baselines and where evidence-chain connection is directly tested. The selector analysis shows that answers chosen by the final evidence-grounded selection step are substantially more accurate than non-selected answers on both datasets. These results support EvidenceFirst as an auditable pipeline with useful internal confidence signals, rather than a method that uniformly dominates every question type.

2Wiki exposes canonicalization pressure. The 2Wiki results show a split between EM and F1. EvidenceFirst v6 improves EM through reader-full context and canonical answer selection, but LightRAG achieves higher F1. This suggests that some EvidenceFirst errors are near misses or overly canonicalized outputs. The next revision should inspect cases where EvidenceFirst loses F1 but not EM, and cases where LightRAG has high F1 without exact match.

Ablations support a narrower component claim. The controlled ablations do not support a simple monotonic story in which every EvidenceFirst component independently improves answer EM/F1. The strongest accuracy component is local/full reader context. Answer refinement is useful on HotpotQA but is not uniformly beneficial on 2Wiki F1. Verification and repair remain central to the method's traceability claim because they expose missing entities, disconnected evidence, bridge gaps, and repair attempts before generation, but their current answer-level gains are mixed and should be presented as diagnostic behavior rather than guaranteed accuracy improvement.

HotpotQA uses one final model. The HotpotQA section reports final EvidenceFirst v6 as the single main model. Earlier HotpotQA development attempts are not part of the reported method comparison or ablation evidence.

HopRAG 2Wiki remains important. The current 2Wiki comparison lacks HopRAG strict, which is the slowest baseline because it builds a graph over 3346 unique passages before generation. When the run finishes, it should be inserted into Table 4. Until then, claims about graph-baseline dominance on 2Wiki should remain cautious.

6 Limitations

This draft has four limitations. First, HopRAG strict on 2Wiki is still running, so the 2Wiki table is incomplete. Second, MS GraphRAG was intentionally deferred on 2Wiki, so the current 2Wiki comparison omits that baseline. Third, the current ablations are system-level switches, and some components interact: reader context can compensate for imperfect graph verification, while answer canonicalization changes EM and F1 differently. Fourth, the current evaluation is answer-level. Because 2Wiki contains gold evidence triples, a stronger version of the paper should evaluate evidence-chain recovery directly, not only final answer EM/F1.

7 Conclusion

We introduced EvidenceFirst, a KG-RAG pipeline that verifies evidence-chain completeness before generation. The method changes the reliability object from the final answer string to the retrieved evidence graph, enabling graph-structural diagnostics such as missing entities, disconnected pairs, bridge gaps, and fallback cases. Current results show that EvidenceFirst is competitive on HotpotQA-1000 and obtains the best completed EM on 2Wiki-500. More importantly, it provides a traceable account of where evidence succeeds or fails, moving KG-RAG toward systems that are not only accurate, but auditable.

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